

ANIMAL KINGDOM

When we look around, we find animals with great *diversity in structure* and form. Since more than a million animal species have been described till now, their *classification becomes essential*. Classification helps in grouping organisms based on common features and also provides a systematic position to newly described species.

Basis of Classification

— 1. Level of Organisation

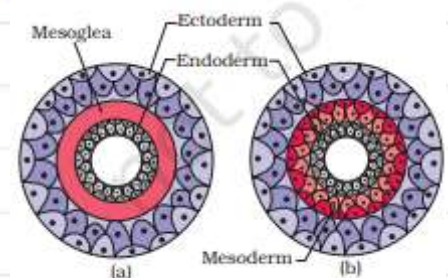
- Cellular → Porifera
- Tissue → Cnidaria
- Organ system → Platyhelminthes onwards

— 2. Body Symmetry

- Asymmetrical → Porifera
- Radial → Cnidaria, Adult Echinodermata
- Bilateral → Platyhelminthes to Chordates

— 3. Germinal Layers

- Diploblastic → Cnidaria
- Triploblastic → Platyhelminthes onwards



— 4. Coelom

- Acoelomate → Platyhelminthes
- Pseudocoelomate → Aschelminthes
- Coelomate → Annelida onwards

— 5. Segmentation

- Segmented → Annelida, Arthropoda, Chordata
- Unsegmented → Others

— 6. Notochord

- Present → Chordates
- Absent → Non-chordates

Levels of Organisation

All members of Animalia are multicellular, but show different levels of organisation.

Cellular level:

Cells loosely arranged, no tissues.

Example: Sponges (Porifera).

Tissue level:

Cells performing similar functions form tissues.

Example: Cnidaria.

Organ level:

Tissues grouped to form organs specialized for specific functions.

Example: Platyhelminthes.

Organ system level:

Organs work together to form organ systems.

Found in Annelida, Arthropoda, Mollusca, Echinodermata, Chordata.

Digestive system:

Incomplete → one opening (mouth = anus), e.g., Platyhelminthes.

Complete → two openings (mouth + anus).

Circulatory system:

Open type → blood pumped into body cavity, directly bathes tissues.

Closed type → blood flows inside vessels (arteries, veins, capillaries).

Symmetry

Asymmetrical: Body cannot be divided into equal halves (e.g., Sponges).

Radial symmetry: Any plane through central axis divides body into equal halves (e.g., Cnidaria, Ctenophora, adult Echinodermata).

Bilateral symmetry: Only one plane divides body into identical left & right halves (e.g., Annelids, Arthropods, Chordates).

Diploblastic and Triploblastic Organisation

Based on germ layers:

Diploblastic:

Two layers – Ectoderm (outer) and Endoderm (inner).

Between them → Mesoglea (undifferentiated layer).

Example: Cnidaria.

Triploblastic:

Three layers – Ectoderm, Mesoderm, Endoderm.

Example: Platyhelminthes to Chordates.

Coelom (Body Cavity)

Space between body wall and gut wall.

Types:

Acoelomates:

No body cavity.

Example: Platyhelminthes.

Pseudocoelomates:

Body cavity not fully lined by mesoderm, mesoderm scattered as pouches.

Example: Aschelminthes.

Coelomates (True coelom):

Body cavity fully lined by mesoderm.

Example: Annelida, Mollusca, Arthropoda, Echinodermata, Hemichordata, Chordata.

Segmentation (Metamerism)

Body externally & internally divided into segments with repetition of organs.

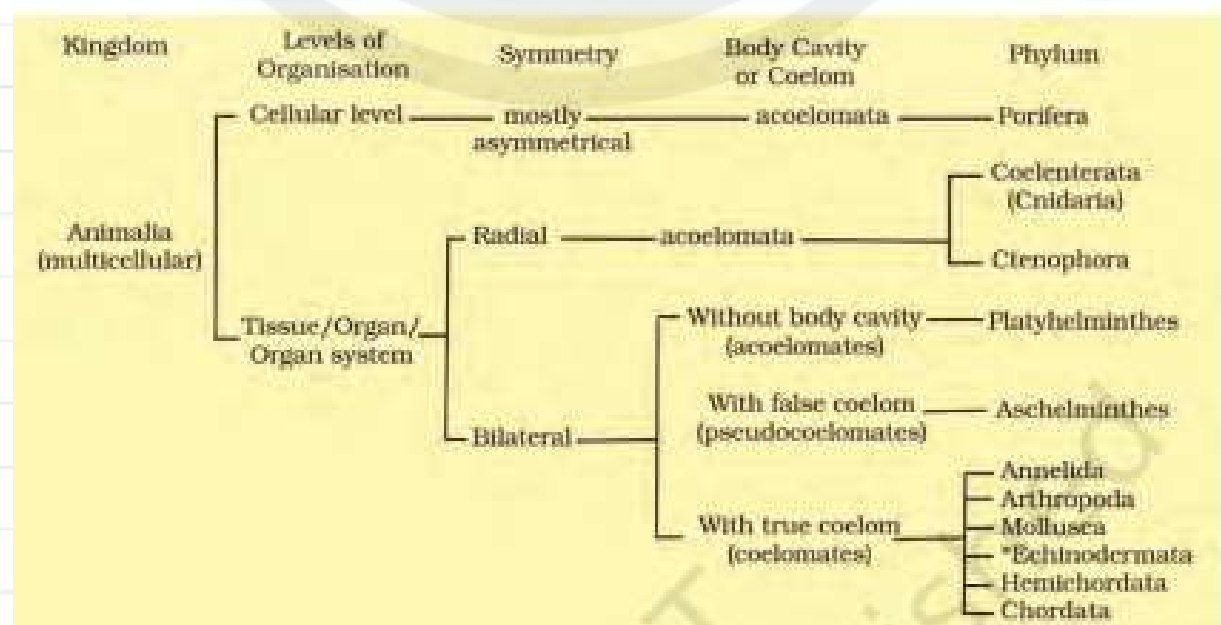
Example: Earthworm (Annelida) → shows metameric segmentation.

Notochord

A mesodermally derived rod-like structure on dorsal side during embryonic development.

Chordates: Notochord present.

Non-chordates: Notochord absent (from Porifera to Echinodermata).



Coelomates (True coelom):

Body cavity fully lined by mesoderm.

Example: Annelida, Mollusca, Arthropoda, Echinodermata, Hemichordata, Chordata.

Segmentation (Metamerism)

Metameric Segmentation (True segmentation)

Body divided into repeated segments.

Each segment has a repetition of some organs.

Example: Earthworm (Annelida).

This phenomenon is called Metamerism.

Pseudometamerism (Superficial segmentation)

External segmentation present, but internal organs not repeated.

Example: Tapeworm (Platyhelminthes).

No segmentation

Body not divided into segments.

Example: Roundworm (Ascaris), sponges, molluscs. *Example:* Earthworm (Annelida) → shows metameric segmentation.

Notochord

A mesodermally derived rod-like structure on dorsal side during embryonic development.

Chordates: Notochord present. Animals with notochord (at least in embryonic stage).

Non-chordates: Notochord absent (from Porifera to Echinodermata).

Animal Kingdom

- |
- | — Porifera → Sponges, canal system, intracellular digestion
- | — Cnidaria (Coelenterata) → Hydra, corals, cnidoblasts, polyp & medusa, metagenesis
- | — Ctenophora → Comb jellies, 8 comb plates, bioluminescence
- | — Platyhelminthes → Flatworms, acoelomate, flame cells, parasitic
- | — Aschelminthes (Nematoda) → Roundworms, pseudocoelomate, dioecious
- | — Annelida → Earthworm, leech, metameric segmentation, closed circulation
- | — Arthropoda → Insects, jointed appendages, chitinous exoskeleton, open circulation
- | — Mollusca → Snail, oyster, octopus, radula, mantle & shell
- | — Echinodermata → Starfish, water vascular system, larva bilateral, adult radial
- | — Hemichordata → Balanoglossus, worm-like, stomochord, open circulation
- | — Chordates (With notochord, dorsal nerve cord, pharyngeal gill slits, post-anal tail)

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|— Urochordata (Tunicata) → Notochord in larval tail (e.g., Ascidia)

|— Cephalochordata → Notochord throughout life (e.g., Amphioxus)

|— Vertebrata → Notochord replaced by vertebral column

- Ventral muscular heart
- Kidneys, paired appendages
- Classes: Pisces, Amphibia, Reptilia, Aves, Mammalia

Phylum – Porifera (Sponges)

Mostly marine, primitive multicellular, asymmetrical animals.

Cellular level of organisation.

Canal system → water enters through ostia → spongocoel → exits via osculum.

Functions: food gathering, respiration, excretion.

Choanocytes (collar cells) line the canals.

Intracellular digestion.

Skeleton: spicules or spongin fibres.

Hermaphrodite (eggs & sperms in same individual).

Reproduction:

Asexual → fragmentation.

Sexual → internal fertilisation, indirect development (larva).

Examples: Sycon (Scypha), Spongilla, Euspongia.

Phylum – Coelenterata (Cnidaria)

Aquatic, mostly marine, radially symmetrical.

Cnidoblasts (nematocysts) on tentacles → stinging, defence, prey capture.

Tissue level of organisation, diploblastic.

Gastrovascular cavity with single opening (mouth).

Digestion: extra + intracellular.

Corals: skeleton of CaCO_3 .

Two forms:

Polyp (sessile, cylindrical: Hydra, Adamsia).

Medusa (umbrella-shaped, free-swimming: Aurelia).

Metagenesis → alternation of generations (Obelia).

Examples: Physalia, Adamsia, Pennatula, Gorgonia, Meandrina.

Phylum – Ctenophora (Comb jellies)

Exclusively marine, radial symmetry, diploblastic.

8 rows of ciliated comb plates → locomotion.

Digestion: extra + intracellular.

Show bioluminescence.

Hermaphrodite, sexual reproduction.

Fertilisation external, development indirect.

Examples: Pleurobrachia, Ctenoplana.

Phylum – Platyhelminthes (Flatworms)

Dorso-ventrally flattened body.

Mostly endoparasites.

Bilateral symmetry, triploblastic, acoelomate, organ level.

Hooks & suckers in parasitic forms.

Some absorb nutrients directly through body surface.

Flame cells → excretion & osmoregulation.

Hermaphrodite, internal fertilisation.

Development → multiple larval stages.

Planaria shows regeneration.

Examples: Taenia (Tapeworm), Fasciola (Liver fluke).

Phylum – Aschelminthes (Roundworms)

Body circular in cross-section (roundworms).

Free-living/aquatic/terrestrial or parasitic.

Organ-system level, bilateral, triploblastic, pseudocoelomate.

Complete alimentary canal with muscular pharynx.

Excretory tube → wastes via pore.

Dioecious (sexes separate); females larger.

Fertilisation internal, development direct or indirect.

Examples: Ascaris, Wuchereria (filarial worm), Ancylostoma (hookworm).

Phylum – Annelida (Segmented worms)

Aquatic/terrestrial, free-living or parasitic.

Organ-system, bilateral, triploblastic, coelomate, metamerically segmented.

Locomotion: longitudinal + circular muscles; parapodia in aquatic forms (Nereis).

Closed circulatory system.

Nephridia → excretion & osmoregulation.

Nervous system: paired ganglia + double ventral nerve cord.

Nereis → dioecious; earthworm & leech → monoecious.

Examples: *Nereis*, *Pheretima* (earthworm), *Hirudinaria* (leech).

Phylum – Arthropoda

Largest phylum (>2/3 of all species).

Organ-system, bilateral, triploblastic, coelomate, segmented.

Exoskeleton of chitin.

Body: head, thorax, abdomen.

Jointed appendages.

Respiration: gills, book gills, book lungs, or tracheae.

Open circulation.

Excretion: Malpighian tubules.

Sense organs: antennae, eyes, statocysts.

Mostly dioecious, fertilisation internal, oviparous.

Development: direct/indirect.

Examples:

Apis (honey bee), *Bombyx* (silkworm), *Laccifer* (lac insect).

Vectors: *Anopheles*, *Culex*, *Aedes*.

Locusta (locust), *Limulus* (king crab – living fossil).

Phylum – Mollusca

Second largest phylum.

Aquatic/terrestrial.

Organ-system, bilateral, triploblastic, coelomate.

Body unsegmented, soft, calcareous shell.

Body parts: head-foot, visceral hump, mantle.

Mantle cavity with gills → respiration & excretion.

Mouth with radula (rasping organ).

Dioecious, oviparous, indirect development.

Examples: Pila, Pinctada, Sepia, Loligo, Octopus, Aplysia, Dentalium, Chiton.

Phylum – Echinodermata

Marine, spiny body with endoskeleton of calcareous ossicles.

Organ-system, triploblastic, coelomate.

Adults → radial symmetry, larvae → bilateral symmetry.

Digestive system complete (mouth ventral, anus dorsal).

Water vascular system → locomotion, food capture, respiration.

No excretory system.

Dioecious, sexual reproduction, external fertilisation.

Development indirect (larva).

Examples: Asterias, Echinus, Antedon, Cucumaria, Ophiura.

Phylum – Hemichordata

Earlier subphylum of Chordata, now separate phylum (non-chordates).

Worm-like, marine, organ-system level, bilateral, triploblastic, coelomate.

Body: proboscis, collar, trunk.

Stomochord (rudimentary notochord-like structure).

Circulation: open.

Respiration: gills.

Excretion: proboscis gland.

Dioecious, external fertilisation, indirect development.

Examples: Balanoglossus, Saccoglossus.

Phylum – Chordata

Key features:

Notochord

Dorsal hollow nerve cord

Pharyngeal gill slits

Post-anal tail

Organ-system, bilateral, triploblastic, coelomate.

Closed circulatory system.

Subphyla:

Urochordata (Tunicata) → notochord only in larval tail. (Ex: Ascidia, Salpa, Doliolum).

Cephalochordata → notochord throughout life, head to tail. (Ex: Amphioxus / Branchiostoma).

Vertebrata → notochord in embryo, replaced by vertebral column.



1. Subphylum Urochordata (Tunicata)

Exclusively marine animals.

Mostly sessile, some are free-floating.

Body is enclosed in a tunic (tough cellulose-like covering).

Notochord: Present only in the larval tail region; disappears in the adult.

Nerve cord: Present in larva, reduced in adult.

Adults are mostly sedentary filter-feeders.

Examples: *Ascidia*, *Salpa*, *Doliolum*.

2. Subphylum Cephalochordata

Small, fish-like marine animals.

Notochord extends from head to tail and persists throughout life.

Transparent and laterally compressed body.

Dorsal tubular nerve cord is present.

Respiration by gill-slits.

Show features of chordates more clearly than Urochordates.

Example: *Branchiostoma* (commonly called *Amphioxus* or *Lancelet*).

Protochordates (Urochordata + Cephalochordata)

Both are exclusively marine.

Primitive chordates.

Lack a vertebral column, cranium, and advanced organ systems.
Bridge between invertebrates and vertebrates.

3. Subphylum Vertebrata (Craniata)

Highly advanced chordates.

During embryonic development, a notochord is present, but in adults it is replaced by a vertebral column (backbone) made of cartilage or bone.

Distinct cranium (brain box) present → hence also called Craniata.

Possess well-developed systems:

Circulatory system: Ventral muscular heart (2, 3, or 4 chambers), closed circulation.

Excretory system: Kidneys for excretion and osmoregulation.

Paired appendages: Fins or limbs.

Endoskeleton: Cartilaginous or bony.

Nervous system: Well-developed brain and sensory organs.

◆ Important Note:

All vertebrates are chordates (since they possess notochord at least in embryonic stage).

But all chordates are not vertebrates (since protochordates lack vertebral column).

Division of Vertebrata

Agnatha (jawless vertebrates)

Only class: Cyclostomata

Gnathostomata (jawed vertebrates)

Superclass Pisces (fishes – bear fins)

Class Chondrichthyes (cartilaginous fishes)

Class Osteichthyes (bony fishes)

Superclass Tetrapoda (bear limbs)

Class Amphibia

Class Reptilia

Class Aves

Class Mammalia

Classes of Vertebrata

1. Class Cyclostomata (Agnatha)

Jawless, ectoparasites on fishes.

Body elongated, no paired fins or scales.

6-15 pairs of gill slits for respiration.

Circular, sucking mouth without jaws.

Cranium + vertebral column cartilaginous.

Marine, but migrate to freshwater for spawning; die after spawning.

Class Cyclostomata (Agnatha) : - Examples: Petromyzon (Lamprey), Myxine (Hagfish).

2. Class Chondrichthyes (Cartilaginous fishes)

Marine, streamlined body.

Endoskeleton cartilaginous.

Notochord persistent.

Mouth ventral.

5-7 pairs of gill slits, without operculum.

Skin with placoid scales; teeth modified placoid scales.

Powerful jaws, predaceous.

No air bladder → must swim continuously.

Heart 2-chambered.

Males have claspers; internal fertilisation.

Mostly viviparous.

Examples: Scoliodon (Dogfish), Pristis (Sawfish), Carcharodon (Great white shark), Trygon (Sting ray).

3. Class Osteichthyes (Bony fishes)

Marine & freshwater fishes.

Endoskeleton bony.

Streamlined body, terminal mouth.

4 pairs of gills covered by operculum.

Skin with cycloid/ctenoid scales.

Class Cyclostomata (Agnatha) : - Examples: Petromyzon (Lamprey), Myxine (Hagfish).

Air bladder present (for buoyancy).

Heart 2-chambered.

Cold-blooded (poikilothermous).

External fertilisation, oviparous, direct development.

Examples:

Marine: Exocoetus (Flying fish), Hippocampus (Seahorse).

Freshwater: Labeo (Rohu), Catla (Katla), Clarias (Magur).

Aquarium: Betta (Fighting fish), Pterophyllum (Angel fish).

4. Class Amphibia

Dual life" – live in water & on land.

Body: head + trunk, tail may/may not be present.

Two pairs of limbs.

Moist, scaleless skin.

Eyelids present, tympanum = ear.

Cloaca present (common chamber for excretion + reproduction).

Respiration by gills, lungs & skin.

Heart 3-chambered.

Cold-blooded.

Fertilisation external, oviparous, indirect development.

Class Amphibia :- Examples: Rana (Frog), Bufo (Toad), Hyla (Tree frog), Salamandra (Salamander), Ichthyophis (Limbless amphibia).

5. Class Reptilia

Creeping/crawling animals.

Body covered with dry, cornified skin (scales/scutes).

Tympanum present, no external ear.

Limbs 2 pairs (when present).

Heart usually 3-chambered; 4-chambered in crocodiles.

Cold-blooded (poikilothermous).

Fertilisation internal; oviparous; direct development.

Examples: Chelone (Turtle), Testudo (Tortoise), Crocodilus (Crocodile), Chameleon (Tree lizard), Naja (Cobra), Bangarus (Krait), Vipera (Viper).

6. Class Aves (Birds)

Forelimbs → wings (flight).

Body covered with feathers.

Beak present, no teeth.

Hindlimbs with scales (for walking, swimming, perching).

Skin dry, no glands (except oil gland at tail base).

Endoskeleton bony; long bones hollow (pneumatic).

Additional digestive chambers: crop & gizzard.

Additional digestive chambers: crop & gizzard.

Heart 4-chambered.

Warm-blooded (homoiothermous).

Respiration via lungs + air sacs.

Internal fertilisation, oviparous, direct development.

Examples: *Corvus* (Crow), *Columba* (Pigeon), *Psittacula* (Parrot), *Struthio* (Ostrich), *Pavo* (Peacock), *Aptenodytes* (Penguin).

7. Class Mammalia

Widely distributed – land, air, water.

Distinct mammary glands → nourish young ones.

Body covered with hair.

Two pairs of limbs, adapted to walking, swimming, flying, burrowing etc.

External ears (pinnae).

Teeth of different types (heterodont).

Heart 4-chambered.

Warm-blooded (homoiothermous).

Respiration by lungs.

Internal fertilisation; mostly viviparous; direct development.

Class Mammalia :- Examples:

Oviparous: Ornithorhynchus (Platypus).

Viviparous: Macropus (Kangaroo), Camelus (Camel), Canis (Dog), Panthera tigris (Tiger), Balaenoptera (Blue whale), Homo sapiens (Humans).

